Introduction to Mathematics and Optimization

January, 2023

PROBLEM 1

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function given by

$$f(x, y) = 3x + 2y$$

and $C \subseteq \mathbb{R}^2$ the subset given by

$$C = \left\{(x, y) \in \mathbb{R}^2 \,\middle|\, x + y \leq 10,\ 2x + y \leq 15,\ x \geq 0,\ y \geq 0\right\}.$$

We consider the two optimization problems

$$\begin{aligned} &\max f(x) \\ &x \in C \end{aligned} \tag{1}$$

and

$$\begin{aligned} &\min\ -f(x) \\ &x \in C \end{aligned} \tag{2}$$

(a) Explain what the relationship is between the optimization problems (1) and (2).

(b) Sketch the set C and prove without reference to the sketch that C is a convex subset of $\mathbb{R}^2$.

(c) Show that C is a compact subset of $\mathbb{R}^2$.

(d) Which result in the curriculum ensures that the optimization problems (1) and (2) can be solved, and why?

(e) Solve the optimization problem (1) using Fourier-Motzkin elimination.

(f) Show that the KKT conditions corresponding to (2) have a solution.

PROBLEM 2

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function given by

$$f(x, y) = x^2 - 4xy + 4y^2 - 6x - 3y$$

and $C \subseteq \mathbb{R}^2$ the subset

$$C = \left\{ (x, y) \in \mathbb{R}^2 \,\middle|\, x + y \leq 10,\ 2x + y \leq 15,\ x \geq 0,\ y \geq 0 \right\}.$$

We consider the optimization problem

$$\begin{aligned} &\min f(x) \\ &x \in C. \end{aligned} \tag{3}$$

(a) Show that

$$f(x, y) = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -6 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

(b) Show that f is a convex function.

(c) Write down the KKT conditions for (3).

(d) Show using the KKT conditions that $(5, 4)$ cannot be an optimal solution to (3).

(e) Use the KKT conditions to find a solution to (3). It may be assumed without proof that any optimal solution (x, y) to (3) has $x > 0$ and $y > 0$.

(f) Is there more than one solution to (3)?

(g) Use a previous part to find a vector $v \in \mathbb{R}^2 \setminus \{0\}$ such that $f(tv) = tf(v)$ for $t \in \mathbb{R}$. Is f strictly convex?

PROBLEM 3

Consider the system of linear equations

$$\begin{array}{rcrcrcl} x & + & y & + & 2z & = & 1 \\ x & + & 2y & + & 3z & = & 1 \\ & & y & + & 2z & = & t \end{array} \tag{4}$$

where $t \in \mathbb{R}$.

(a) The system of equations (4) can be written in matrix form $Av = b$, where

$$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \qquad \text{and} \qquad b = \begin{pmatrix} 1 \\ 1 \\ t \end{pmatrix}.$$

Find the matrix A.

(b) Show that (4) can be solved uniquely for all $t \in \mathbb{R}$.

(c) Show that

$$A^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix},$$

where A is the matrix from (a).

(d) Let $f : \mathbb{R}^4 \to \mathbb{R}$ be given by

$$f(x, y, z, t) = (1 - x - y - 2z)^2 + (1 - x - 2y - 3z)^2 + (t - y - 2z)^2.$$

Find the global minima of f. Is f strictly convex?